

GROUP TWO

A Guess Number Game

A command-line number guessing game in Python



The Problem

We built a command-line number guessing game in Python. Here is one round:

The computer picks a secret number (1–100 by default)



The player enters a guess



The program answers “Too low”, “Too high” or “Correct”

LEVEL 1

Basic

One guess. The program compares it and shows the result.

LEVEL 2

Intermediate

A loop keeps asking until the guess is correct and counts attempts.

LEVEL 3

Problem Solving

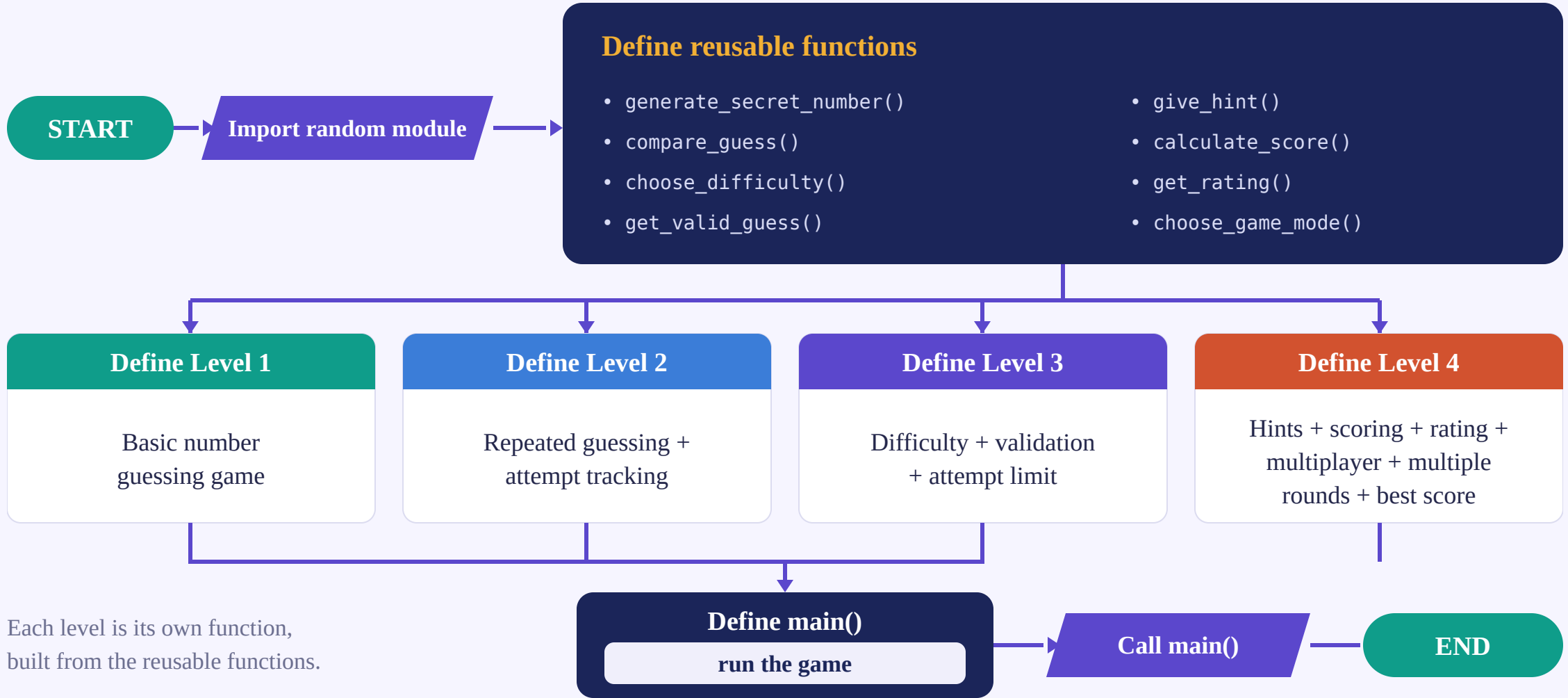
Easy, Medium or Hard, input validation and a limit on attempts.

LEVEL 4

Challenge

Hints, score, star rating, multiplayer, many rounds and a best score.

The Flowchart



The Reusable Functions

Function	Parameters	Returns
<code>generate_secret_number</code>	low=1, high=100	A random integer
<code>compare_guess</code>	guess, secret_number	"Too low", "Too high" or "Correct"
<code>choose_difficulty</code>	none	low, high, max_attempts
<code>get_valid_guess</code>	low, high	A valid integer guess
<code>give_hint</code>	secret_number, low, high	A hint (even/odd or range)
<code>calculate_score</code>	max_attempts, attempts	$(\text{max} - \text{attempts} + 1) \times 100$
<code>get_rating</code>	attempts	★★★★ / ★★★ / ★
<code>choose_game_mode</code>	none	"1" or "2"

Each function has one job and sends its result back to the caller.

Code Walkthrough

Explain: input → logic → output

`generate_secret_number()`

Generate a random integer between low and high, then return it.

`choose_difficulty()`

Ask for Easy, Medium, or Hard; map the choice to low, high, and max_attempts.

`give_hint()`

Use the secret number to produce a hint, such as even/odd or a narrowed range.

`get_rating()`

Use the number of attempts to determine the star rating and return the stars.

`compare_guess()`

Compare the guess with the secret number and return Too low, Too high, or Correct.

`get_valid_guess()`

Keep asking until the input is an integer within the allowed range; return the valid guess.

`calculate_score()`

Use max_attempts and attempts to calculate the score: $(\text{max_attempts} - \text{attempts} + 1) \times 100$.

`choose_game_mode()`

Keep asking until the player enters 1 or 2, then return the selected mode.

Level 1 to Level 4 — Code Walkthrough

Brief algorithm for the logic inside each level

level1() Basic

- Generate a secret number.
- Take one guess from the player.
- Compare the guess with the secret number.
- Display the result.

One guess → one comparison → result

level2() Intermediate

- Generate a secret number.
- Keep asking for guesses with a loop.
- Count each attempt.
- Stop when the guess is correct.

Loop → guess → compare → count → repeat

level3() Problem Solving

- Choose a difficulty level.
- Get a valid guess using input validation.
- Track attempts against the maximum allowed.
- End with a win or Game Over.

Difficulty → validation → attempt limit → result

level4() Challenge

- Run the guessing game with hints.
- Give a hint after every 3 wrong guesses.
- Calculate the score and star rating.
- Support multiplayer, multiple rounds and best score.

Hints → scoring → rating → modes → rounds

Technical Concepts Used

Functional programming

Organizing code around small functions that perform clear tasks and return results.

Code reuse

Writing a function once and using it again in different parts of the program.

Separation of concerns

Dividing a program into parts so each part handles a different responsibility.

Single responsibility

Giving each function one clear job instead of making one function do many jobs.

Default parameters

Preset values used when an argument is not provided, such as low=1 and high=100.

Unpacking

Assigning multiple returned values directly to separate variables.

Also used: random module · while loops · conditionals · try/except

Live Demo

1. Level 3: choose difficulty, enter invalid input, win or run out of attempts
2. Level 4 single player: hints after 3 wrong guesses, score and stars
3. Level 4 multiplayer: two players, one secret number
4. Play another round: the best score updates

Questions?

Thank you · Group Two